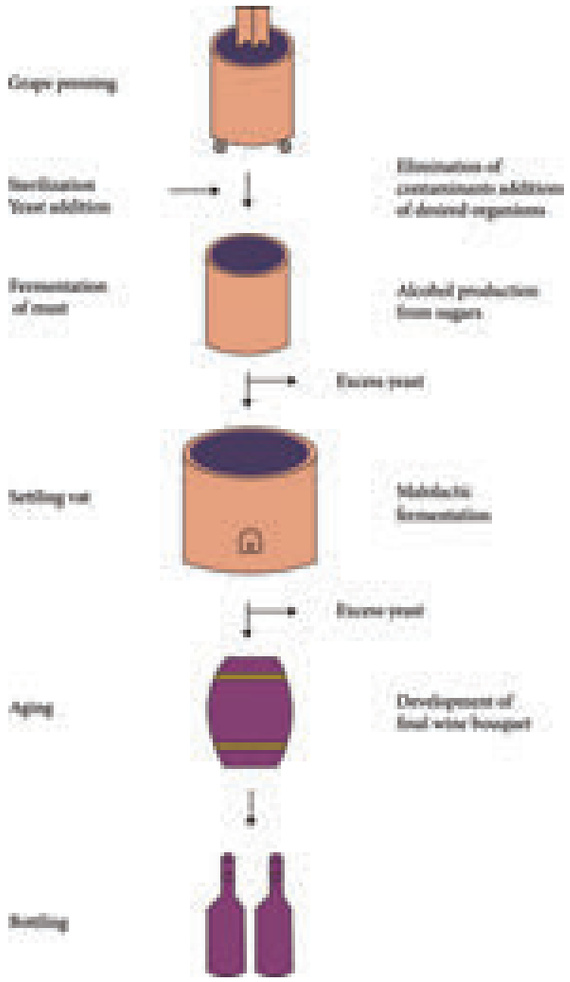


**Figure 6.11:** Steps involved in wine production

**Infobits**

**Zymology:**

Zymology also known as "Zymurgy" (from the Greek–means, the workings of fermentation) is an applied science which studies the biochemical process of fermentation and its practical uses. It includes selection of fermenting yeast and bacteria, species and their use in brewing, wine making, fermenting milk and other fermented foods. The wine yeast Zymurgist one who studies or practices zymurgy; a knowledgeable brewer.

## 6.8 Industrial Production of Single Cell Protein

Single cell protein refers to the microbial cells or total protein extracted from pure microbial cell culture (monoculture) which can be used as protein supplement for humans or animals. During ancient times, the tribes in the Central African Republic used a spiral shaped Cyanobacterium named *Spirulina platensis* as food. They collected it as mats from the bottom of seasonally dried up ponds and shallow waters around Lake Chad and dried them in the sun and made small cakes called "Dihe".

During the World war II, when there were shortage in proteins and vitamins in the diet, the Germans produced yeasts and a mould named *Geotrichum candidum* was used as food.

The term Single Cell Protein was coined by C.L Wilson (1966) at Massachusetts Institute of Technology (MIT), to represent the cells of algae, bacteria, yeasts and fungi, grown for their protein contents. The name was introduced by Prof. Scrimshow of MIT in 1967. The organisms like *Pseudomonas facilis*, P. *flava, Chlorella, Anabaena, Spirulina, Chlamydomonas*, and *Agaricus* are commonly used for SCP production. Large scale production of SCP is shown in the Figure 6.12

There are several methods available for SCP production. In the Japanese method, flat tray is used with artificial sunlight algae are cultivated in shallow ponds with mechanical stirrers or in deeper ponds (not more than 20–30 cm deep) with circulation pumps. Optimum, light is an important parameter for maximum



TN_GOVT_XII_Micro_Biology_CH06.indd 106 27-02-2019 14:46:21